ANSWER KEY

SAT Practice Test 7 - Standardized Edition

Section 1: Reading and Writing

Module 1 **|** Module 2 Easier **|** Module 2 Harder

Section 2: Math

Module 1 **|** Module 2 Easier **|** Module 2 Harder

Answers and explanations are preserved from the original source file.

Original test content retained. Module order and presentation standardized to the supplied reference structure.

Section 1: Reading and Writing

Module 1 (27 Questions) - All Students

Time: 32 minutes

Question 1

Answer: A) Molds

Ka “shapes” clay into faces, meaning she physically forms or molds the clay. Therefore, “Molds” is correct.
“Grows” (B), “Designs” (C), and “Positions” (D) do not match the meaning of “shapes” in this context.

Question 2

Answer: B) imitate

The Alhambra's mosaic follows the decorative conventions of Moroccan zellige. “Imitate” means to follow or copy the style of. “Replicate” (A) means to make an exact copy, which is too strong. “Disregard” (C) means to ignore. “Surpass” (D) means to exceed.

Question 3

Answer: C) curbing

The sentence describes how excess inventory piles up and is hard to clear. The blank needs a word meaning “controlling” or “restraining.” “Curbing” means to restrain. “Distributing” (A) means spreading. “Cataloging” (B) means recording. “Concealing” (D) means hiding.

Question 4

Answer: A) trademark

The tiered rooftop is nearly universal in pagodas — a “trademark” is a distinguishing, characteristic feature. “Contradiction” (B) implies conflict. “Inspiration” (C) means a source of ideas. “Replica” (D) means a copy.

Question 5

Answer: B) It provides a foundational description of a phenomenon that is central to the discussion that follows.

The underlined portion defines parallel adaptation, setting up the entire discussion. It is not an example of research (A), not correcting a misunderstanding (C), and not introducing a method (D).

Question 6

Answer: C) It presents some specific words in Quechua, describes the general linguistic phenomenon exemplified by those words, and then states that this phenomenon occurs frequently in Quechua.

The text starts with specific words (wasi, wasiwasi), then explains reduplication, and finally says there are numerous examples in Quechua.

Question 7

Answer: C) Text 1 discusses a single plausible cause of Boreas, whereas Text 2 evaluates two possible causes.

Text 1 presents evidence for a celestial impact (one cause). Text 2 examines volcanic caldera collapse and salt tectonics (two alternatives) and rules them out.

Question 8

Answer: D) David anticipates having the opportunity to be introduced to both the sisters and their mother.

David “was hoping not only to observe our seminar but to seek an introduction to the professor herself at Mother's residence.” He wanted to meet both the sisters and their mother.

Question 9

Answer: A) Because of the manner in which they were created, they likely have visual qualities that are regarded as more typical of Carolina Naturalist paintings than the qualities in works by Reeves are.

Walsh adopted Palmer's “rapid layering,” which created the impressionistic style synonymous with the group. Reeves worked with greater precision, making her work less typical of the group's style.

Question 10

Answer: D) It should be understood as just one of several factors that influence cycling activity in Copenhagen.

The passage states that “many factors influence people's decision-making about whether to cycle” and “none of these factors in isolation fully explains cycling habits.”

Question 11

Answer: C) Argentina reported 210 damaging mite species, which is more than either Chile or Brazil reported.

The conclusion states Argentina had more mites. The table shows: Argentina 210, Chile 31, Brazil 12. Choice C directly states this comparison.

Question 12

Answer: B) "When ye weep for lands across the foam, / The stranger in his distant plight, / Remember those who starve

at home / And perish in your city's night."

The claim says the speaker criticizes those who champion distant causes while neglecting local suffering. Choice B directly addresses this contrast between distant and local suffering.

Question 13

Answer: C) 120 leaf pairs show opposite-direction asymmetry, whereas approximately 50 pairs show same-direction asymmetry.

From the graph, sugar maple (dark gray) shows ~120 pairs for opposite-direction and ~50 for same-direction, supporting the claim that opposite-direction is far more common.

Question 14

Answer: A) Although both species tended to extend their flowering period later into autumn in years with late frost onset, the extension was much greater for *H. hispanica* than for *H. non-scripta*.

This directly shows the invasive species gained a measurably bigger advantage from delayed frost, supporting the conclusion.

Question 15

Answer: A) being recruited to participate in a study by someone with whom one is personally connected may create a sense of commitment to persist with participation in the study.

Personal connection creates a feeling of obligation, so participants recruited through personal networks are more likely to stay in the study.

Question 16

Answer: B) hangs

“Currently” indicates present tense. The painting currently hangs at the museum. “Hangs” is present tense and agrees with the singular subject.

Question 17

Answer: C) bears

The subject “The flexor pollicis brevis” is singular and requires present tense “bears.” The sentence describes a general anatomical fact.

Question 18

Answer: D) began

A specific historical event in 1868 requires simple past tense. The other options create fragments or don't work grammatically.

Question 19

Answer: D) rings, a

“With over two millennia of growth data in its rings” is an introductory prepositional phrase (dependent), not an independent clause. A comma correctly separates it from the main clause “a single tree like this...can reveal the ecological history.” A semicolon (B) is incorrect because semicolons require two independent clauses, and the first part cannot stand alone as a sentence. Choice A incorrectly capitalizes “A.” Choice C creates a nonsensical compound.

Question 20

Answer: B) Congress:

A colon correctly introduces the list of names that follows. A semicolon (A) doesn't introduce lists. A period (C) would create a fragment.

Question 21

Answer: C) Indeed,

“Indeed” reinforces and elaborates on the previous point that the pen name “accomplished far more than simply disguising his identity.”

Question 22

Answer: A) Ultimately,

The passage describes conflicting feelings that are resolved: beauty and harshness are inseparable. “Ultimately” signals this final resolution.

Question 23

Answer: C) The wandering albatross and the Andean condor can both be found in the Southern Hemisphere.

Both birds live in the Southern Hemisphere (Andes = South America; Southern Ocean). This states a shared geographic trait.

Question 24

Answer: D) Classified as Beaufort number 7, a near gale produces wind speeds of 50–61 km/h.

Directly and concisely states the near gale's wind speed, which is the goal.

Question 25

Answer: B) Langston Hughes's first published poetry collection was called *The Weary Blues*.

Directly names the title, which is the goal.

Question 26

Answer: A) The stereognostic bag activity in the Montessori method and form drawing in Waldorf education are both exercises that develop children's spatial skills.

Directly states what both exercises share — developing spatial skills.

Question 27

Answer: B) With *Equilibrium*, Meireles joins the generations of artists who have helped sustain the mobile as an artistic medium since its resurgence in the 2000s.

Places the 2020 sculpture within the historical timeline of mobiles since their resurgence in the 2000s.

Section 1: Reading and Writing

Module 2 - Easier Path (27 Questions)

Time: 32 minutes

Question 1

Answer: C) Draws

Tom “traces” letters on his slate with chalk — he draws them by moving the chalk to form letter shapes.

Question 2

Answer: B) relevant to

Past environmental pressures may also be “relevant to” (applicable to) today's threatened species.

Question 3

Answer: D) commended

The reviewers praised both films with positive language. “Commended” means praised.

Question 4

Answer: A) To discuss an ongoing experiment

Bauer's experiment from 1885 is still being conducted — vials continue to be opened and checked.

Question 5

Answer: D) It provides details about Calderon's artworks, discusses specific examples of her work, and relates them to one of her artistic goals.

The text describes Calderon's method, gives two examples, and connects both to her goal of showing resilience and connectedness.

Question 6

Answer: B) It provides examples of realizations the speaker has gradually accepted.

The underlined part gives examples of what the speaker has come to accept through experience.

Question 7

Answer: C) A city's culinary scene can benefit from the city being named a Creative City of Gastronomy.

Both texts agree the designation CAN benefit a city's food scene, though Text 2 adds that marketing is needed to fully realize benefits.

Question 8

Answer: B) The El Hierro giant lizard is an example of a species unique to the Canary Islands archipelago that is highly specialized and therefore particularly susceptible to habitat disturbances.

Captures both main points: high specialization and resulting vulnerability.

Question 9

Answer: A) Its location may be considered remote.

Earth artists “typically install large-scale works outdoors, often in remote locations.” The Mojave Desert qualifies as remote.

Question 10

Answer: B) 1,000,000 units sold compared to the approximately 15,800,000 units sold of the Game Boy.

Uses the correct Palm Pilot number (1,000,000) compared to the much larger Game Boy number (15,800,000).

Question 11

Answer: D) Monarch butterflies, painted lady butterflies, and red admiral butterflies display navigation shifts of more than 15 degrees in response to both ground-level and overhead magnetic fields.

Directly contradicts the assumption that butterflies do not detect overhead fields.

Question 12

Answer: A) "The color is repellant...but now I find myself tracing its convolutions with growing intensity."

Shows progression from “idle curiosity” to “growing intensity” — increasing fixation.

Question 13

Answer: C) Argentina, because its rate of dairy exports to the US increased in the post-FTA period relative to the pre-FTA period, while its rate of total dairy exports decreased during the same period.

Trade redirection means exports to the FTA partner increase while total exports decrease (no new production, just shifting where goods go). Argentina's data shows this pattern: US exports grew from 10.1% to 17.2%, while total exports fell from 39.5% to 3.6%.

Question 14

Answer: D) across cultural contexts, people who share similar profiles of personality traits tend to prefer listening to

similar types of music.

The study found consistent personality-music correlations across 41 countries, suggesting personality drives music preference universally.

Question 15

Answer: B) winter flounder and the mako shark are at least partly determined by the ambient water temperature.

Both ectotherms and the mako shark show body temperatures influenced by ambient water temperature.

Question 16

Answer: B) will likely continue

Future prediction requires future tense: “will likely continue.”

Question 17

Answer: D) the 3rd century, these

“Between the 8th century and the 3rd century” is a prepositional phrase, not an independent clause. A period (B) would create a sentence fragment. A comma correctly separates this introductory phrase from the main clause “these deities appeared in epic poems.” Choice A creates a subordinate clause with no main clause. Choice C incorrectly implies causation.

Question 18

Answer: A) display

“Baskets” is plural; present tense “display” agrees with it and the parallel structure (“are woven...and display”).

Question 19

Answer: B) researcher Elena Bianchi,

“Researcher” is a title before the name — no comma between title and name. A comma after the name closes the introductory phrase.

Question 20

Answer: C) sheds

Singular subject “it” in present tense requires “sheds.”

Question 21

Answer: C) volume, respectively;

Comma before “respectively,” then semicolon to join two independent clauses.

Question 22

Answer: A) With this in mind,

Connects the artist's environmental concern to her subsequent action of using natural dyes.

Question 23

Answer: D) When shown three versions...8.7% of participants chose a modified version.

Directly states the 8.7% figure, emphasizing modified-version selection.

Question 24

Answer: A) Although Darjeeling and chamomile are both types of tea, the former's astringent flavor is more intense.

Directly contrasts the two teas' astringency levels.

Question 25

Answer: C) Previously believed to be extinct, a living population of *Dryococelus australis* was identified on Lord Howe Island.

Includes the location (Lord Howe Island), which is the goal.

Question 26

Answer: C) Like ornithologist Florence Merriam Bailey, ethologist Nikolaas Tinbergen dedicated a scientific career to the study of animal life.

Both studied animal life — captures the shared trait.

Question 27

Answer: B) Compressive heating is the process by which an object increases in temperature when it is compressed.

Gives a clear definition of the concept.

Section 1: Reading and Writing

Module 2 - Harder Path (27 Questions)

Time: 32 minutes

Question 1
Answer: D) Find
The merchant was trying to "locate" (find) Raskolnikov among the crowd.

Question 2
Answer: B) germane to
"Germane to" means relevant or applicable to — drawing a parallel between ancient and modern challenges.

Question 3
Answer: D) praised
The critics described works with positive language. "Praised" means spoke favorably of. "Lionized" (A) is too strong.

Question 4
Answer: A) To describe a researcher's alternative interpretation of a cognitive phenomenon and indicate how it contrasts with an established model
Presents Kowalczyk's alternative theory and contrasts it with the "blocking" model.

Question 5
Answer: D) It provides details about Kamoda's artistic methods, discusses contrasting examples of his work, and connects them to one of his artistic principles.
Describes technique, gives contrasting examples, and connects both to "tension between control and surrender."

Question 6
Answer: B) It extends the catalog of tropical items that evoke the speaker's distant homeland.
Continues the list of tropical fruits, evoking the speaker's Caribbean homeland.

Question 7
Answer: A) Text 1 presents pilot program data as sufficient evidence for a conclusion about UBI's effects, whereas Text 2 questions whether pilot data can support such a broad conclusion.
Text 1 uses pilot data to conclude; Text 2 argues pilots are too limited to generalize.

Question 8
Answer: A) A commonly used research organism may yield less broadly applicable results than researchers have assumed, because laboratory populations have diverged from their wild counterparts.
Captures Petrov's argument about lab fruit fly divergence limiting generalizability.

Question 9
Answer: A) the process of actively discussing and agreeing upon domestic responsibilities may matter more to marital satisfaction than the specific outcome of that discussion.
Couples who negotiated had the highest satisfaction regardless of the outcome — the process matters more than the result.

Question 10
Answer: B) It requires readers to consider the social dynamics between the writer and the intended recipient of the letter.
Okafor argues readers must evaluate who the letter is addressed to and what rhetorical purpose it serves.

Question 11
Answer: C) Queensland reported 185 damaging aphid species, which is more than either Victoria or Western Australia reported.
Directly compares Queensland's 185 aphids to Victoria's 14 and Western Australia's 38.

Question 12
Answer: A) "The tower that men built with pride and care / Now hosts the ivy's quiet reign..."
Illustrates human ambition (building a tower) versus nature's indifference (ivy taking over, builders forgotten).

Question 13
Answer: C) 175 plant pairs show discordant asymmetry, whereas approximately 70 pairs show concordant asymmetry.
From the graph, big bluestem shows ~175 discordant and ~70 concordant pairs.

Question 14
Answer: A) Although both species extended their active feeding period in years with late freeze-up, the extension was

substantially greater for perch than for char.

Directly shows perch gained a bigger advantage from delayed freeze-up.

Question 15

Answer: A) studying material in varied environments across sessions should produce better retention than studying in the same environment each time.

If contextual variation creates more retrieval pathways, then varying environments should enhance the effect.

Question 16

Answer: B) will likely continue

Future prediction in present/future context requires “will likely continue.”

Question 17

Answer: D) the 1st century BCE, these

Same pattern as S1M2E Q17. “Between the 15th century BCE and the 1st century BCE” is a prepositional phrase (dependent). A period (B) would create a fragment. A comma correctly separates it from the main clause.

Question 18

Answer: A) showcase

Plural subject “prints” in present tense = “showcase.”

Question 19

Answer: B) researcher David Okonkwo,

Title before name, no comma between title and name. Comma after name closes the introductory phrase.

Question 20

Answer: C) drops

Singular subject “it,” present tense, habitual action = “drops.”

Question 21

Answer: C) height, respectively;

Comma before “respectively,” semicolon between independent clauses.

Question 22

Answer: A) Ultimately,

Resolves the tension between forgetting and remembering — both are inescapable.

Question 23

Answer: D) When shown four versions...27.6% of participants chose an altered version.

Emphasizes the 27.6% who chose altered versions, matching the goal.

Question 24

Answer: A) Although English mustard and Dijon mustard are both prepared condiments, the former's isothiocyanate pungency is more intense.

Contrasts the two mustards' pungency levels.

Question 25

Answer: B) A living *Metasequoia glyptostroboides*, previously known only from fossils, was discovered in 1943.

Specifies when (1943) the species was found alive.

Question 26

Answer: C) Like mycologist Beatrix Potter, lichenologist Elke Mackenzie devoted her scientific career to the study of organisms in the fungal kingdom.

Both studied organisms in the fungal kingdom.

Question 27

Answer: A) Indeed,

Reinforces and expands on how Arendt's phrase accomplished far more than description.

Section 2: Math

Module 1 (22 Questions) - All Students

Time: 35 minutes

Question 1
Answer: B) x: −1, 0, 1 → y: 1, 3, 5
The line passes through (−1, 1), (0, 3), and (1, 5). Equation: $y = 2x + 3$. Verify each point.

Question 2
Answer: 80
Line of best fit at $x = 4$ gives approximately 80 cm.

Question 3
Answer: B) 9
$8x + 24 = 8(x + 3) = 72$, so $x + 3 = 9$.

Question 4
Answer: D) $0 < n < 420$
Beam length is positive ($n > 0$) and less than 420 ($n < 420$). Combined: $0 < n < 420$.

Question 5
Answer: A) 13
Third side = 31 − 7 − 11 = 13 meters.

Question 6
Answer: D) 10
Mean = (9 + 6 + 14 + 11 + 10) ÷ 5 = 50 ÷ 5 = 10.

Question 7
Answer: B) It is plausible that the actual mean is between 6.8 and 9.6.
Confidence interval: 8.2 ± 1.4 = [6.8, 9.6].

Question 8
Answer: B) $-11x^6 + 7x^9 - 7x + 8$
Distribute the negative sign and combine like terms: $-4x - 3x = -7x$; 6 + 2 = 8.

Question 9
Answer: D) A line with a moderate negative slope roughly bisecting the data cloud.
Weak negative correlation requires a negatively-sloped line through the middle of the data.

Question 10
Answer: 6,840
$g(40) = 180(40 - 2) = 180 \times 38 = 6{,}840$ degrees.

Question 11
Answer: A) $h(x) = 3^x + 12$
Check: $h(0) = 1 + 12 = 13$, $h(1) = 3 + 12 = 15$, $h(2) = 9 + 12 = 21$. All match.

Question 12
Answer: 2, 7, or 13
x-intercepts where $f(x) = 0$: $x = 2$, $x = 7$, or $x = 13$.

Question 13
Answer: C) 28
From eq 2: $y = 5 - 3x$. Substitute into eq 1: $x + 5(5 - 3x) = 23 \rightarrow -14x = -2 \rightarrow x = 1/7$. Then $y = 32/7$. So $4x + 6y = 4/7 + 192/7 = 196/7 = 28$.

Question 14
Answer: D) No additional information is needed.
Both triangles have angles 52°, 93°, and 35° (since angles sum to 180°). By AA similarity, two matching angle pairs suffice.

Question 15
Answer: A) −6
From equation 2: $y = x + 1$. Substitute into equation 1: $x^2 + x + 1 = 31 \rightarrow x^2 + x - 30 = 0 \rightarrow (x + 6)(x - 5) = 0$. Solutions: $x = -6$

or $x = 5$. Answer: A) -6.

Question 16

Answer: 5

Dividing $f(x)$ by $(x - 13)$ removes the x-intercept at $x = 13$ (creating a hole). So g has $6 - 1 = 5$ x-intercepts.

Question 17

Answer: C) 16

Numerator = 0: $x = 5$ or $x = 11$. Neither makes denominator zero. Sum = $5 + 11 = 16$.

Question 18

Answer: 190

Line through origin parallel to $y = 38x + 7$ has equation $y = 38x$. At $x = 5$: $d = 190$.

Question 19

Answer: D) 20

Arc length = $(90/360) \times C = C/4 = 5 \rightarrow C = 20$ inches.

Question 20

Answer: A) 48.01

Jupiter weight = 236.4% of 150 = 354.6 lbs. $x = (170.25/354.6) \times 100 \approx 48.01\%$.

Question 21

Answer: 9

Substitute $y = 14$: Eq 1 gives $q + 14w = 196$; Eq 2 gives $3q - 14w = 84$. Adding: $4q = 280$, so $q = 70$. Then $14w = 196 - 70 = 126$, so $w = 9$.

Question 22

Answer: tan 25° ≈ 0.4663

In the right triangle, $\tan B$ = opposite/adjacent = AC/BC. Since angle B = 25°, $\tan 25° \approx 0.4663$.

Section 2: Math

Module 2 - Easier Path (22 Questions)

Time: 35 minutes

Question 1
Answer: B) 18
$p(6) = 24 - 6 = 18$.

Question 2
Answer: C) The maximum estimated profit, in dollars
Downward-opening parabola; vertex is the highest point = maximum profit.

Question 3
Answer: A) $y = 8x + 70$
Slope = 8, y-intercept = 70.

Question 4
Answer: D) 289π
Area = $\pi r^2 = \pi(17^2) = 289\pi$.

Question 5
Answer: B) $c = p^2/10 + 36/10$
Square both sides: $p^2 = 10c - 36$. Add 36: $p^2 + 36 = 10c$. Divide by 10: $c = (p^2 + 36)/10$.

Question 6
Answer: A) −76
$f(x) = -76x$ is in the form $y = mx$; slope = −76.

Question 7
Answer: 236
Slope = $(205 - 48)/(2015 - 1995) = 7.85$K/year. $x = 205 + 7.85(4) \approx 236$.

Question 8
Answer: D) $-6 + 2\sqrt{14}$
Using the quadratic formula: $w = (-12 \pm \sqrt{(144+80)})/2 = (-12 \pm \sqrt{224})/2 = (-12 \pm 4\sqrt{14})/2 = -6 \pm 2\sqrt{14}$. Choice D matches one solution.

Question 9
Answer: 13
Since $RG \parallel HI$, angle FRG = angle FHI = 68° (corresponding angles). In triangle FHI: $p + 68 + 31 = 180$, so $p = 81$. Therefore $p - r = 81 - 68 = 13$.

Question 10
Answer: C) $f(x) = 185(1.50)^{x/3}$
Starting at 185, increasing by 50% means multiplying by 1.50. After x days = $x/3$ periods of 3 days.

Question 11
Answer: D) −12
From the graph and boundary line, $s = -12$ produces the correct boundary and shaded region.

Question 12
Answer: C) Infinitely many
Both sides simplify to $72x - 108$, which is true for all x.

Question 13
Answer: D) 55
After 12 months: value multiplied by 0.45, retaining 45% and losing 55%. So $p = 55$.

Question 14
Answer: C) $p(w) = 750 - 150w$
After 3 weeks at 100/week: $800 - 300 = 500$ aphids. Population at week w: $500 - 150(w-3) = 950 - 150w$. Still needed to reach 200: $(950 - 150w) - 200 = 750 - 150w$.

Question 15

Answer: D) $3x - 4y = -12$

Line through (0,3) and (−4,0). Slope = 3/4. Equation: $y = (3/4)x + 3$. Multiply by 4: $3x - 4y = -12$. Verify: $3(-4) - 4(0) = -12$ ✓; $3(0) - 4(3) = -12$ ✓.

Question 16

Answer: B) $5x + 20$

Factor: 25x^2(x+8) - 25(x+8)^3 = 25(x+8)[x^2 - (x+8)^2] = -400(x+8)(x+4). Therefore, 5(x+4) = 5x + 20 is a factor, so B is correct.

Question 17

Answer: D) $x = 50y - 100$

Rewrite: $x = 62.5y - 100$. For no solution, the second equation must have the same slope ratio but different constant when both equations are in standard form.

Question 18

Answer: B) 0.73

$f(s+1) = ab^{s+1} = ab^s \cdot b = t \cdot b$. Given $f(s+1) = 0.73t$: $b = 0.73$.

Question 19

Answer: C) 190.19

$0.72\ m^3 = 720{,}000\ cm^3$. Convert: $720{,}000 \div 3{,}785.41 \approx 190.19$ gallons.

Question 20

Answer: 335

$g(x) = 3(x-5)^2 - 8$. Y-intercept: $g(0) = 3(25) - 8 = 67$. For $6g(x)$: y-intercept = $6(67) = 402$. Difference = $402 - 67 = 335$.

Question 21

Answer: −17

Discriminant < 0: $c^2 - 324 < 0 \rightarrow c^2 < 324 \rightarrow -18 < c < 18$. Least integer: −17.

Question 22

Answer: 900

Let N = starting number. Prey at end: $N + 31N = 32N$. Predators: $N + 2.2N = 3.2N$. $p = (32N - 3.2N)/3.2N \times 100 = 28.8/3.2 \times 100 = 900$.

Section 2: Math

Module 2 - Harder Path (22 Questions)

Time: 35 minutes

Question 1
Answer: C) 27
$h(5) = 32 - 5 = 27$.

Question 2
Answer: B) The minimum estimated cost, in dollars
Upward-opening parabola; vertex is the lowest point = minimum cost.

Question 3
Answer: A) $y = 12x + 45$
Slope = 12, y-intercept = 45.

Question 4
Answer: B) $c = p^2/14 + 50/14$
Square both sides: $p^2 = 14c - 50$. Then $c = (p^2 + 50)/14$.

Question 5
Answer: A) −54
$f(x) = -54x$ is in the form $y = mx$; slope = −54.

Question 6
Answer: 265
Slope = $(231 - 63)/20 = 8.4$K/year. $x = 231 + 8.4(4) = 264.6 \approx 265$.

Question 7
Answer: D) $-5 + 8 = 3$
$w = (-10 \pm \sqrt{256})/2 = (-10 \pm 16)/2$. Solutions: $w = 3$ or $w = -13$. Choice D: $-5 + 8 = 3$ ✓.

Question 8
Answer: 7
$MN \parallel GH$: angle EMN = angle EGH (corresponding), so $b = 72°$. Triangle EGH: $a + 72 + 29 = 180 \rightarrow a = 79$. $a - b = 7$.

Question 9
Answer: C) $f(x) = 310(1.40)^{x/4}$
Starting at 310, increasing 40% means multiplying by 1.40. After x days = x/4 periods.

Question 10
Answer: D) −18
From the graph, $t = -18$ produces the correct boundary and shading.

Question 11
Answer: C) Infinitely many
Both sides simplify to $75x - 125$. True for all x.

Question 12
Answer: D) 72
After 12 months: value multiplied by 0.28, retaining 28%, losing 72%. $p = 72$.

Question 13
Answer: C) $p(w) = 1{,}680 - 160w$
After 3 weeks at +100/week: 900 + 300 = 1,200 bees. Population at week w: $1{,}200 + 160(w-3) = 720 + 160w$. Still needed: $2{,}400 - (720 + 160w) = 1{,}680 - 160w$.

Question 14
Answer: D) $7x - 4y = 28$
Line through (4,0) and (0,−7). Slope = 7/4. Verify: $7(4) - 4(0) = 28$ ✓; $7(0) - 4(-7) = 28$ ✓.

Question 15
Answer: A) -14x - 42
Factor the expression: $49x^2(x+6) - 49(x+6)^3 = 49(x+6)[x^2 - (x+6)^2] = -588(x+6)(x+3)$. Therefore, $-14(x+3) = -14x - 42$ is a factor, so A is correct.

Question 16
Answer: D) $x = (175/3)y - 350/3$
Rewrite: $x = (175/3)y - (350/3)$. For no solution, the second equation must have the same coefficient ratio but different constant.

Question 17
Answer: B) 0.91
$f(s+1) = tb = 0.91t$, so $b = 0.91$.

Question 18
Answer: C) 250.96
$0.95\ m^3 = 950{,}000\ cm^3$. $950{,}000 \div 3{,}785.41 \approx 250.96$ gallons.

Question 19
Answer: 51
$g(x) = 3(x-3)^2 - 10$. $g(0) = 27 - 10 = 17$. For $4g(x)$: y-intercept $= 4(17) = 68$. Difference $= 68 - 17 = 51$.

Question 20
Answer: −13
Discriminant < 0: $d^2 - 196 < 0 \rightarrow d^2 < 196 \rightarrow -14 < d < 14$. Least integer: -13.

Question 21
Answer: 660
Prey: $N + 18N = 19N$. Predators: $N + 1.5N = 2.5N$. $p = (19 - 2.5)/2.5 \times 100 = 660$.

Question 22
Answer: 20
Substitute $y = 11$: $q + 11w = 300$ and $4q - 11w = 100$. Adding: $5q = 400$, $q = 80$. Then $11w = 300 - 80 = 220$, $w = 20$.

ANSWER SUMMARY — QUICK REFERENCE

S1 M1: 1.A 2.B 3.C 4.A 5.B 6.C 7.C 8.D 9.A 10.D 11.C 12.B 13.A 14.A 15.A 16.B 17.C 18.D **19.D** 20.B 21.A 22.A 23.C 24.D 25.B 26.A 27.B

S1 M2 Easier: 1.C 2.B 3.D 4.A 5.D 6.B 7.C 8.B 9.A 10.B 11.D 12.A **13.C** 14.D 15.B 16.B **17.D** 18.A 19.B 20.C 21.C 22.A 23.D 24.A 25.C 26.C 27.B

S1 M2 Harder: 1.D 2.B 3.D 4.A 5.D 6.B 7.A 8.A 9.A 10.B 11.C 12.A 13.A 14.A 15.A 16.B **17.D** 18.A 19.B 20.C 21.C 22.A 23.D 24.A 25.B 26.C 27.A

S2 M1: 1.B 2.80 3.B 4.D 5.A 6.B 7.B 8.B 9.B 10.6840 11.A 12.2/7/13 13.C 14.D **15.A** 16.5 17.C 18.190 19.D **20.A 21.9** 22.≈0.47

S2 M2 Easier: 1.B 2.C 3.A 4.D 5.B 6.A 7.236 **8.D** 9.13 10.C 11.D 12.C 13.D **14.C** 15.D 16.B 17.D 18.B **19.C** 20.335 21.−17 **22.900**

S2 M2 Harder: 1.C 2.B 3.A 4.B 5.A 6.265 7.D 8.7 9.C 10.D 11.C 12.D **13.C** 14.A 15.A 16.D 17.B 18.C 19.51 20.−13 21.660 **22.20**

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